

MUHAMMED CAN ARICA

Computer Engineering Student | Unity & C# Game Development | Gameplay Programming

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PROFILE

I am a fourth-year Computer Engineering student at Izmir Bakircay University, focused on game development with Unity and C#. I build gameplay mechanics, reusable game systems and playable 2D prototypes, with an emphasis on testing, iteration and solving technical problems through clear software structures.

EDUCATION

Izmir Bakircay University

B.Sc. Computer Engineering - 4th Year Student

Menemen, Izmir

2023 - Expected Graduation: 2027

PROJECT EXPERIENCE

Wimmia

2D Action / Platform Gameplay Prototype - Personal Project

Unity, C#

Apr 2026 - Present

- Developing a playable 2D gameplay prototype with player movement, combat/damage systems, enemy behavior, camera logic, room transitions and environmental interactions.
- Built boss encounters with multiple attack patterns, area hazards, weak points and attack timing, together with gameplay feedback such as hit stop, knockback and camera shake.
- Separated movement, combat, camera, enemy and boss responsibilities into independent systems to make mechanics easier to test, modify and reuse.

DungeonWeaver

Seed-Based Procedural Dungeon Generation System

Unity, C#

Jul 2026 - Present

- Built a reusable procedural dungeon system for Unity that generates the same dungeon layout from the same seed, making generation deterministic and reproducible.
- The system automatically generates rooms with different shapes and sizes, room-connection graphs, corridors, doorways, start/boss roles and room encounter flows.
- Separated the generation core from the demo gameplay using Core, Editor and test assemblies, then packaged the reusable core for installation in other Unity projects through a Git URL and Unity Package Manager.

PipeMuzzle

Data-Driven 2D Pipe Puzzle Game Prototype

Unity, C#

Aug 2026 - Present

- Built a playable 2D puzzle prototype where players rotate pipe tiles in 90-degree steps to create a continuous path between a source and a target.
- Represented pipe directions with bit masks and used a BFS-based connectivity check to determine whether a level has been solved.
- Implemented ScriptableObject-based level data, runtime board generation, automatic camera fitting, move tracking, restart and level-progression flows.

INTERNSHIP EXPERIENCE

Eti Krom A.S. - Information Technology Intern

Software Development, Network Systems & Computer Vision

Kovancilar, Elazig

Jul 2026 - Aug 2026

- Developed NetScope/NetworkMonitor with ASP.NET Core/C# and React/TypeScript to monitor device reachability, SNMP data, incidents, notifications, topology and configuration history.
- Developed CamerBound for the site's entry/exit cameras using Python, PySide6, OpenCV, PaddleOCR and OpenVINO to recognize license plates and maintain local vehicle records.
- Gained broad hands-on experience with enterprise networking and network operations, including IP addressing/subnetting, switching/routing, VLANs/trunks, DHCP/DNS, SNMP, firewalls, servers/virtualization and IP cameras; built and tested multiple network topologies and scenarios in Cisco Packet Tracer.

TECHNICAL SKILLS & INTERESTS

- Game Development:** Unity 6, C#, 2D Physics, Cinemachine, Tilemap, Input System, ScriptableObject, gameplay programming, procedural generation, level design
- Software Engineering:** OOP, design patterns, event-driven design, graph/BFS algorithms, REST APIs, Git/GitHub, GitHub Actions, automated testing
- Technologies:** Python, Java, TypeScript, JavaScript, React, ASP.NET Core, SQLite, SignalR, PySide6, OpenCV, PaddleOCR/OpenVINO, MediaPipe, ESP32
- Networking & Tools:** TCP/IP, subnetting, switching, routing, VLAN/trunking, DHCP, DNS, SNMP, ICMP, firewalls, Cisco Packet Tracer, Windows Server, VMware, SecureCRT
- English:** Basic-Intermediate